2026

STRIDE - ASSISTX

GRIPASSIST

Team Name: Brooklyn-07

-Team Members-

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-Problem Statement-

Problem 2: "Adithyan Cannot Hold a Spoon to Eat Independently"

Theme: Oversized Adaptive Utensil Holder for Assisted Feeding (Low-Tech)

Target Users:

Children with Cerebral Palsy, Physical Disabilities, Multiple Disabilities

Proposed Solution

We propose a low-tech oversized adaptive utensil holder designed to improve grip, stability, and ease of use for children with limited hand strength and coordination.

Design Description:

> Orange Part (Utensil Holder): A cylindrical slot designed to securely hold spoons, forks, or toothbrushes. The tight fit prevents slipping and rotation.
> Red Section (Insertion Zone): A guiding channel where the utensil is inserted easily. This ensures quick attachment and removal.
> Blue Section (Palm Support): A broad, curved surface that rests against the palm to distribute pressure and reduce fatigue.
> Yellow Section (Grip Support): A textured or soft region that enhances friction and improves grip stability for the user.

Dimensions: 10–12 cm in length and 3–5 cm in diameter for a child-friendly grip.

Compatibility: Insertion slot fits standard handles (0.5–1.5 cm in diameter).

Materials: 3D-printed using **PLA** (rigid structure) or **TPU** (flexible and comfortable).

Preference: TPU is often preferred for assistive tech due to its soft, rubber-like feel.

Maintenance: Lightweight, reusable, and washable with water and mild soap.

Durability: Designed for repeated daily use while maintaining hygiene.

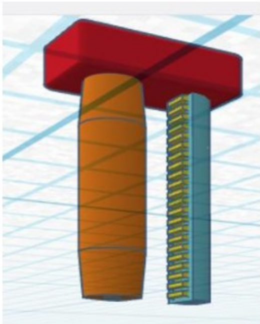
Key Features:

> Oversized ergonomic design for better handling
> Anti-rotation structure to prevent spilling
> Compatible with standard utensils
> Lightweight and easy to carry
> Made from food-safe materials (e.g., silicone, rubber, or foam)

Advantages:

> Improves independence during eating
> Reduces frustration and mess
> Enhances grip strength support
> Easy to clean and maintain
> Low-cost and locally manufacturable

Design Illustration:



Evaluation

1. Does it enable the child to hold and use a spoon without dropping it mid-meal?
Yes. The oversized ergonomic design, combined with palm support and enhanced grip texture, improves stability and control, reducing the chances of the spoon slipping or dropping during use.

2. Is it compatible with standard utensils available in schools and at home?
Yes. The holder is designed with a universal slot that fits most standard spoon, fork, and toothbrush handles, ensuring easy compatibility without needing special utensils.

3. Is it safe, hygienic, and easy to clean for daily mealtime use?
Yes. The device is made using food-safe 3D printing materials such as PLA or TPU, with smooth surfaces that are easy to wash using water and mild soap, making it suitable for regular use.

4. Can it be adopted in school canteens and therapy settings without special equipment?
Yes. It is a simple, low-tech solution that requires no additional tools or setup. It can be quickly attached and removed, making it practical for use in schools and therapy environments.

5. Is it low-cost and locally manufacturable from accessible materials?
Yes. The design can be produced using affordable 3D printing materials and basic printing technology, making it cost-effective and easy to manufacture locally.

Conclusion and Future Scope

The proposed oversized adaptive utensil holder provides a simple yet effective solution to assist children with limited motor control in achieving independent eating. Its ergonomic and stable design directly addresses grip challenges, coordination issues, and utensil control.

Future Scope:

- Introduce adjustable sizes for different age groups
- Use advanced materials for improved durability and hygiene
- Add straps or finger guides for enhanced control
- Customize designs based on individual user needs
- Explore mass production using cost-effective methods such as molding or 3D printing.

With further improvements, this design can be widely adopted in homes, schools, and therapy centers to support inclusive and independent living.